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$[Cp^*RhCl_2]_2$ catalyzed three-component coupling cyclization of 2,3-allenoic acids with 2,3-allenols in the presence of $Cu(OAc)_2{\cdot}H_2O$†

Junjie Fan, Xiaoyan Wu, Chunling Fu and Shengming Ma *

Here we report a $[Cp^*RhCl_2]_2$ catalyzed three-component coupling cyclization of 2,3-allenoic acids and 2,3-allenols in the presence of $Cu(OAc)_2{\cdot}H_2O$, affording furanone derivatives. The synthetic potential has been demonstrated. The mechanism has also been demonstrated by isotopic labeling experiments, unveiling unexpected reaction pathways.

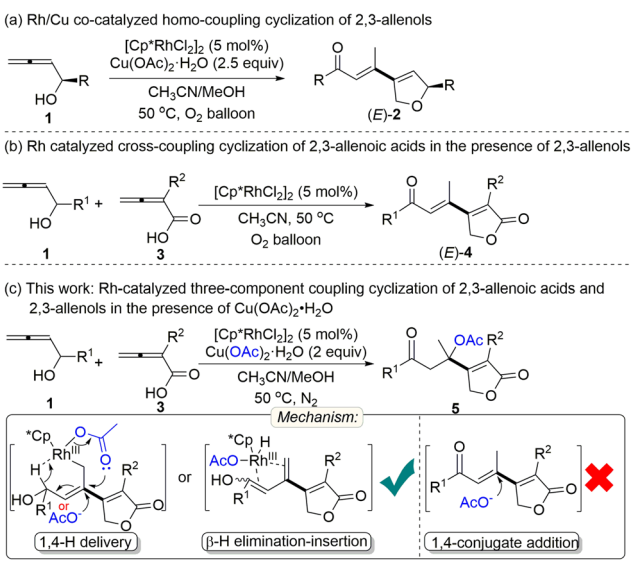
Introduction

Allenes have been developed as versatile building blocks in organic syntheses.[1,2] Our group has developed the Rh/Cu co-catalyzed homo-coupling cyclization of 2,3-allenols **1**, affording a dihydrofuran ring linked with a conjugated enone unit (*E*)-**2** (Scheme 1a).[3] The Rh catalyzed cross-coupling cyclization reaction of 2,3-allenoic acids **3** in the presence of 2,3-allenols **1** affording furan-2(5*H*)-ones (*E*)-**4** has also been reported (Scheme 1b).[4] Here we wish to report a three-component coupling cyclization of 2,3-allenoic acids **3** in the presence of 2,3-allenols **1** and $Cu(OAc)_2{\cdot}H_2O$ catalyzed by a Rh(III) catalyst forming a new type of furan-2(5*H*)-one **5** *via* unique mechanisms (Scheme 1c).

Results and discussion

Initially, the three-component coupling cyclization reaction was tried using 2,3-allenoic acid **3a** (0.2 mmol) and 2,3-allenol **1a** (0.3 mmol) in the presence of $[Cp^*RhCl_2]_2$ (7.5 mol%) and $Cu(OAc)_2{\cdot}H_2O$ (0.6 mmol) in CH_3CN (0.6 mL)/methanol (30 μL) under an oxygen atmosphere (Scheme 2-1).[3] The reaction afforded the homo-coupling product (*E*)-**2a** in 12% NMR yield derived from two molecules of 2,3-allenol **1a**. In addition, two formal 1,4-conjugate addition products of (*E*)-**4aa** with acetate, **5aa** (30% yield by NMR) and **6aa** (8% yield by NMR), were detected. However, interestingly, when (*E*)-**4aa** was prepared on purpose and then treated under the same catalytic conditions with $[Cp^*RhCl_2]_2$ (5 mol%) and $Cu(OAc)_2{\cdot}H_2O$ (2 equiv.) for 12 h, **5aa** was not formed (Scheme 2-2), indicating a unique mechanism for the formation of **5aa** and **6aa**.

To optimize the selectivity, we adjusted the ratio of **3a**/**1a** from 1/1.5 to 1.5/1 and observed that the yield of **5aa** was improved and the formation of (*E*)-**2a** was not observed (Table 1, entries 2 and 3). Interestingly, further increasing the loading of **3a** improved the selectivity of **5aa**/**6aa** (entry 4). The reaction in open air gave similar results (entry 5 *vs.* entry 4). Surprisingly, when the reaction was conducted in an inert atmosphere of N_2, **5aa** was formed exclusively (entry 6). Upon studying the effects of $Cu(OAc)_2{\cdot}H_2O$ loading and concentration (entries 7–10), the conditions described in entry 9 were

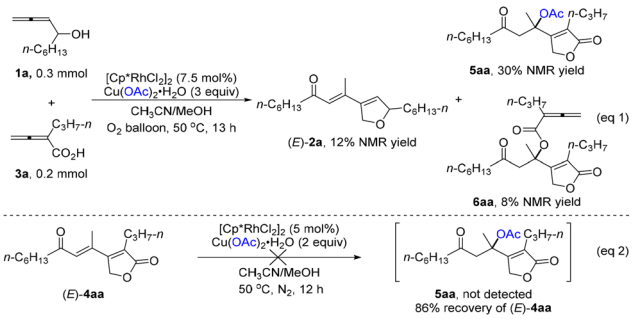


Scheme 1 Rh-catalyzed coupling cyclization involving 2,3-allenols and/or 2,3-allenoic acids.

Laboratory of Molecular Recognition and Synthesis, Department of Chemistry, Zhejiang University, Hangzhou 310027, Zhejiang Province, People's Republic of China. E-mail: masm@sioc.ac.cn

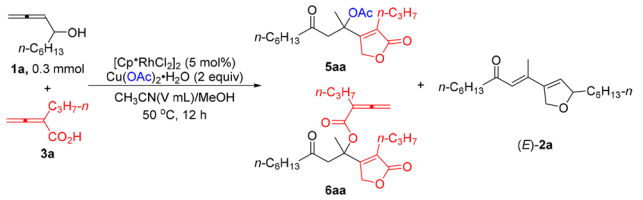
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Scheme 2 The reactions of **1a** with **3a** (eqn (1)) and (*E*)-**4aa** (eqn (2)) in the presence of $[Cp^*RhCl_2]_2$ and $Cu(OAc)_2{\cdot}H_2O$.

Table 1 Optimization studies for cyclizative cross-coupling of 2,3-allenol **1a** with 2,3-allenoic acid **3a** for the selective formation of product **5aa**[a]

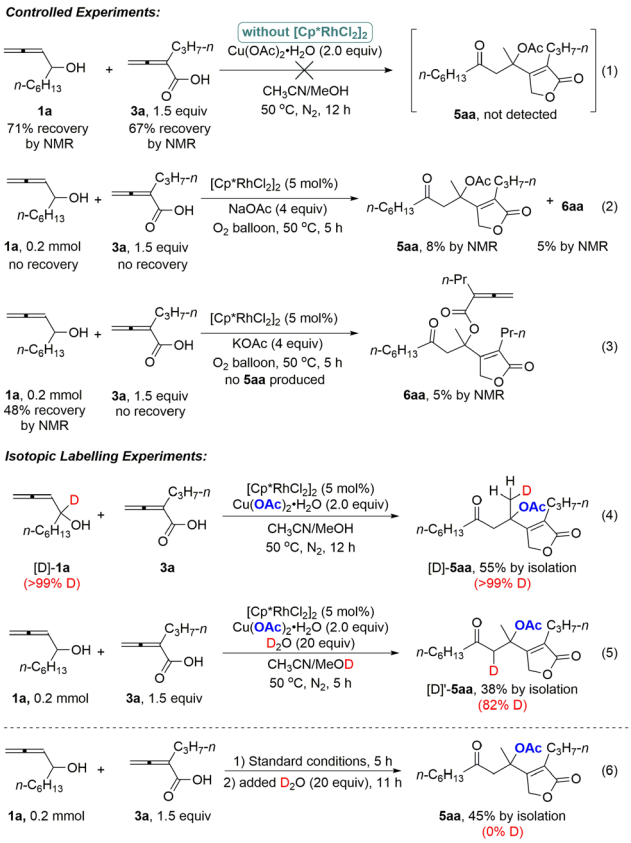


				NMR yield[b] (%)		
Entry	O_2/N_2	*V* (mL)	**3a**/**1a**	**5aa**	**6aa**	(*E*)-**2a**
1[c,d]	O_2	0.6	1/1.5	30	8	12
2[c]	O_2	0.6	1/1	32	28	10
3[e]	O_2	0.6	1.5/1	39	25	0
4	O_2	0.6	2/1	42	13	0
5	Air	0.6	1.5/1	40	16	0
6	N_2	0.6	1.5/1	54	0	0
7[f]	N_2	0.6	1.5/1	50	0	0
8	N_2	0.4	1.5/1	30	0	0
9	N_2	0.8	1.5/1	55	0	0
10	N_2	1.0	1.5/1	49	0	0

[a] The reaction was conducted with 2,3-allenol **1a**, 2,3-allenoic acid **3a**, $[Cp^*RhCl_2]_2$ (5 mol%), and $Cu(OAc)_2{\cdot}H_2O$ (2.0 equiv.) in CH_3CN (0.6 mL)/MeOH (30 μL) at 50 °C for 12 h unless otherwise noted. [b] Determined by NMR analysis of the crude products using mesitylene as the internal standard. [c] 13 h. [d] $[Cp^*RhCl_2]_2$ (7.5 mol%) and $Cu(OAc)_2{\cdot}H_2O$ (3.0 equiv.) were used. [e] 16 h. [f] 2.5 equiv. of $Cu(OAc)_2{\cdot}H_2O$ were used.

chosen as the optimal conditions for the exclusive formation of product **5aa**.

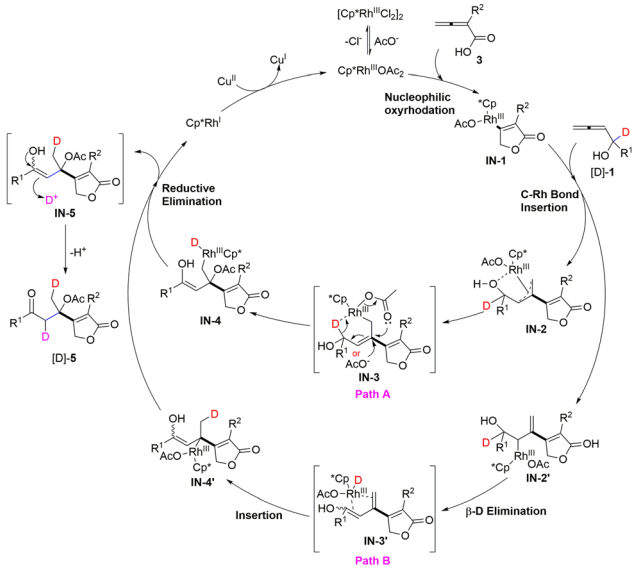
To clarify the mechanism, a control experiment was conducted. In the absence of the rhodium catalyst, the reaction failed to afford **5aa** (Scheme 3-1), indicating that $[Cp^*RhCl_2]_2$ was crucial. In the absence of $Cu(OAc)_2{\cdot}H_2O$ with O_2 as the oxidant and NaOAc as the acetoxylation agent, the reaction afforded **5aa** in only 8% NMR yield and **6aa** in 5% NMR yield (Scheme 3-2), whereas with KOAc as the acetoxylation agent, the reaction afforded **6aa** in 5% NMR yield but not **5aa**, with 48% recovery of **1a** (Scheme 3-3), indicating the importance of $Cu(OAc)_2{\cdot}H_2O$. The reaction of allenoic acid **3a** and deuterated



Scheme 3 Control experiments and isotopic labeling experiments.

allenol [D]-**1a** (>99% D) afforded [D]-**5aa**, with the methyl group connected to the α-carbon atom of acetoxy being monodeuterated (Scheme 3-4). With an additional 20 equivalents of D_2O and using MeOD instead of MeOH, the reaction of **1a** and **3a** afforded the α-deuterated product [D]′-**5aa** (82% D) in 38% yield (Scheme 3-5). When 20 equivalents of D_2O were added after the completion of the reaction of **1a** and **3a** under the standard conditions, **5aa** was isolated in 45% yield without any deuterium incorporation (Scheme 3-6), indicating that [D]′-**5aa** was not produced by **5aa** directly.

Based on these experimental results and the literature,[3–5] two mechanisms were proposed, as shown in Scheme 4. First, $[Cp^*RhCl_2]_2$ would undergo ligand exchange to afford Cp^*RhOAc_2.[6] Then allenoic acid **3** underwent a nucleophilic cyclic oxyrhodation to afford the 2-oxo-2,5-dihydrofuran-4-yl-rhodium intermediate **IN-1**. Subsequently, [D]-**1** underwent insertion into the C–Rh bond to generate the stereo-defined allylic rhodium intermediate **IN-2**. In path A, the coordination of the hydroxyl group would be replaced with that of the C–D bond to form **IN-3**. The subsequent attack on the fully substituted sp^2 carbon center by AcO^-, which was dissociated or coordinated on Rh(III), would induce the migration of the C=C bond and D-delivery to afford **IN-4**. Reductive elimination of **IN-4** would produce Cp*Rh(I) and **IN-5**, which would interact with D_2O to afford [D]-**5**. Finally, the Rh(III) species would be regenerated to complete the catalytic cycle *via* the oxidation of



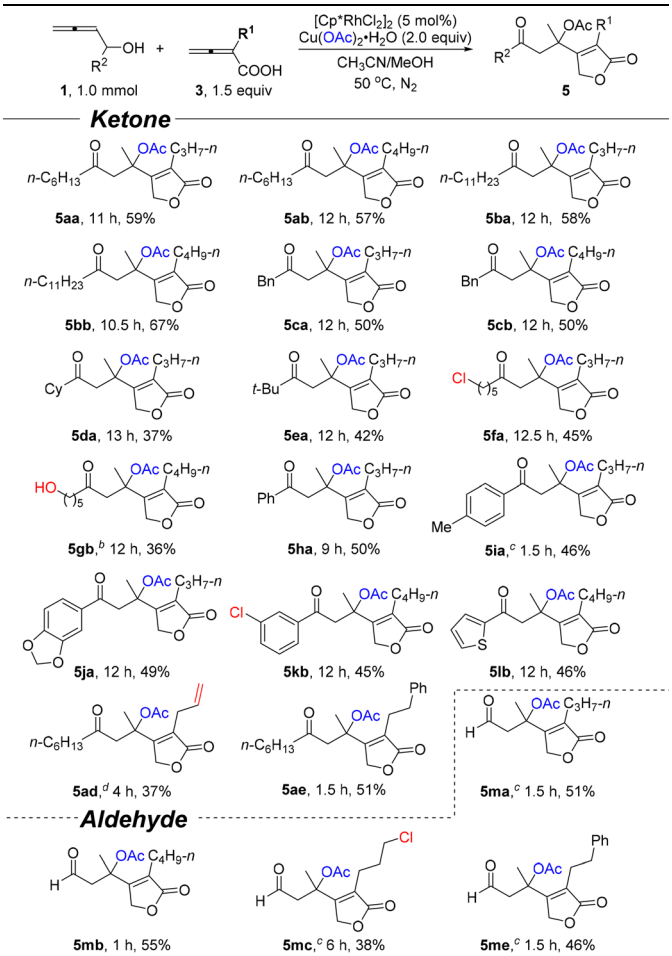
Scheme 4 Proposed mechanism.

Cp*Rh(I) with Cu(II).[7] There is an alternative mechanism (Path B):[8] the allylic rhodium intermediate **IN-2′** may undergo β-D elimination to afford **IN-3′**. Then the terminal C=C bond would insert into the Rh-D bond to produce **IN-4′**. Then the C–OAc moiety of **IN-5** was formed by reductive elimination.

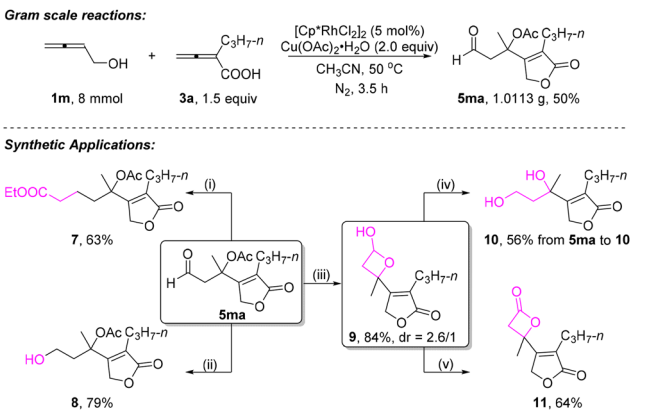
After unveiling the unique mechanism, we explored the substrate scope of this three-component coupling cyclization of 2,3-allenoic acids with 2,3-allenols for the formation of products **5** (Table 2). *n*-Propyl or *n*-butyl substituted allenoic acids were first examined, generating the corresponding products **5aa–5lb** and **5mb** in moderate yields. Allenoic acids equipped with synthetically versatile groups, such as 3-chloropropyl (**3c**), allyl (**3d**), and phenethyl (**3e**), could all react with allenols forming the corresponding products **5ad**, **5ae**, **5mc**, and **5me** in 37%–51% yields. For allenols, R^2 could be 1°-, 2°-, and 3°-alkyl groups, affording **5aa–5ea** in 37%–67% yields. Functional groups such as chlorine (**5fa**) and even free hydroxyl (**5gb**) were also tolerated. R^2 may also be different aryls, affording the corresponding products **5ha–5kb** in 42%–49% yields. Gratifyingly, the heteroaryl substrate **1l** also matched with allenoic acid **3b** affording **5lb** in 46% yield. Simple buta-2,3-dien-1-ol (**1m**) could also work to afford a series of aldehydes **5ma–5me**, leaving new opportunities for further synthetic elaboration.

To further explore the synthetic utility of this methodology (Scheme 5), the reaction was conducted on the 8 mmol scale to afford product **5ma**. **5ma** is synthetically versatile: (i) it reacted with the Wittig reagent to afford the corresponding enoate, which was subsequently hydrogenated with RANEY®-Ni to afford **7** in 63% combined yield in two steps; (ii) the aldehyde unit in **5ma** may be selectively reduced to the primary alcohol **8** with $NaBH_3CN$; (iii) in the presence of triethylamine, a highly strained four-membered ring[9] in **9** may be formed. Subsequent one-pot reduction with $NaBH_4$ afforded 1,3-diol **10**

Table 2 Reaction scope[a]



[a] Standard conditions: $[Cp^*RhCl_2]_2$ (5 mol%), $Cu(OAc)_2{\cdot}H_2O$ (2.0 equiv.), 2,3-allenol **1** (1.0 mmol), and 2,3-allenoic acid **3** (1.5 equiv.) in CH_3CN (2.7 mL)/MeOH (0.13 mL) at 50 °C under a N_2 atmosphere unless otherwise noted. [b] 0.5 mmol scale reaction. [c] Without MeOH. [d] **1a**/**3d** = 1.2/1, $Cu(OAc)_2{\cdot}H_2O$ (3.0 equiv.) was added.



Scheme 5 Gram scale reaction and synthetic applications: (i) step 1: ethyl (triphenylphosphoranylidene)acetate (1.2 equiv.), THF, rt, 37 h; step 2: RANEY®-Ni, MeOH, hydrogen balloon (1 atm), rt, 14 h. (ii) $NaBH_3CN$ (4.5 equiv.), ethyl acetate, rt, 4 h. (iii) Et_3N (6.0 equiv.), MeOH, 85 °C, 24 h. (iv) $NaBH_4$ (2.5 equiv.), rt, 5.5 h. (v) $Fe(NO_3)_3{\cdot}9H_2O$ (10 mol%), TEMPO (10 mol%), KCl (10 mol%), toluene, oxygen atmosphere, 25 °C, 10 h.

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in 56% yield.[10] The hemiacetal unit in **9** may also be oxidized by Fe/TEMPO[11] to afford the potentially useful yet difficult to prepare stained β-lactone **11**[12,13] in 64% yield.

Conclusions

In conclusion, the $[Cp^*RhCl_2]_2$ catalyzed three-component coupling cyclization of 2,3-allenoic acids and 2,3-allenols affording furanone derivatives with an acetate functionality was established. The synthetic potential of the products has been well demonstrated. The reaction proceeded *via* nucleophilic cyclic oxyrhodation, insertion, and 1,4-H delivery promoted by the addition of AcO^- or a β-H elimination–insertion mechanism, as unveiled by mechanistic studies.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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